Colter Snyder

Lakewood, CO · U.S. Citizen · (720) 626-2558 · coltersnyder@gmail.com · in/colter-snyder-a36854183

SUMMARY

Software engineer with two years of development on mission-critical defense simulation software in ITAR- and CUI-controlled Linux environments. Designs and documents object-oriented systems and builds them in C++, Python, and Rust using Git, GitLab CI/CD, test-driven development, and agile practices. Brings graduate study in cyber-physical systems security and hands-on hardware work in FPGA design (Verilog) and 6502/Z80 assembly. M.S. in Computer Science (3.97 GPA); former adjunct instructor to more than 200 students.

EXPERIENCE

Innovative Scientific Solutions, Inc.

Software Engineer II — *Dec 2024 – Present*

- Develop ITAR- and CUI-controlled platforms for probabilistic nuclear weapons effects simulation, covering blast, neutron radiation, thermal fluence, ground shock, cratering, and X-ray effects, for defense and allied government agencies; take part in technical meetings with these customers.
- Work across the full lifecycle on a 4–5 person agile team, leading development of individual features and components: refine requirements, design object-oriented architectures, produce UML (StarUML) and design documentation, implement, and verify with automated tests.
- Spearheaded an internal Python performance-benchmarking tool, a CLI with a GUI front end, that drives our calculation services by individual and combined effect (e.g., overpressure, prompt radiation, or both) and charts the results; also wrote Python test tooling.
- Port legacy physics models from Fortran and Python to Rust, preserving their numerical behavior.
- Build Python (Flask) and Rust back-end services with WebSockets and MongoDB, deployed with Docker on Linux.
- Apply configuration management and secure development practices: Git branching and code review, GitLab CI/CD pipelines, SonarQube static analysis, and strict handling of export-controlled code and data.

Tyler Technologies — Lakewood, CO

Software Development Intern — *May 2023 – Aug 2023*

- Overhauled a 2,000-test automated suite, restoring about 1,500 tests (75%) to passing and giving the team a reliable foundation for test-driven development.
- Migrated a major legacy VB6 desktop application to object-oriented C# on an agile modernization team.

Colorado School of Mines — Golden, CO

Postgraduate Research Assistant, Department of Computer Science — *May 2024 – Dec 2024*

- Developed object detection models for semiconductor cleanroom equipment and integrated them, with Python back-end services, into a Unity AR training app, delivering a working demo in collaboration with the University of Arizona (advisor: Dr. Mehmet Belviranli).

Undergraduate Research Assistant — *Oct 2019 – May 2020*

- Built a Python test suite as one of three developers on *Quality-aware Audio Multicast in Low Power Wireless Networks* (advisor: Dr. Qi Han).

TEACHING & TECHNICAL PRESENTATION

Colorado School of Mines — Golden, CO

Adjunct Professor, Department of Computer Science — *Jan 2023 – Dec 2023*

- Co-taught CSCI 220: Data Structures and Algorithms (C++) to 120 undergraduates, coaching students on object-oriented design, debugging, and algorithmic problem solving (Fall 2023).
- Taught a section of CSCI 102: Intro to Computer Science Lab (Python) to 90 undergraduates (Spring 2023).
- Co-presented a teacher professional development workshop at InnEdCO 2021 as a CS outreach intern.

SKILLS

Languages: C++, C, Python, Rust, C#, Fortran, TypeScript, SQL, Verilog, 6502 & Z80 Assembly

Practices: Object-Oriented Design, Software Architecture & Design Documentation, UML, Agile/Scrum, Test-Driven Development, CI/CD, Configuration Management, Secure Development & Static Analysis

Tools & Platforms: Linux, Git, GitLab CI/CD, SonarQube, Docker, StarUML, MongoDB, WebSockets

EDUCATION

Colorado School of Mines — Golden, CO

Master of Engineering in Nuclear Engineering (in progress) — *Aug 2026 – Expected May 2031*

Master of Science in Computer Science • GPA 3.97 — *Jan 2022 – May 2024*

Bachelor of Science in Computer Science • Magna Cum Laude • GPA 3.77 — *Aug 2019 – Dec 2022*

Relevant coursework: Cyber-Physical Systems Security, IoT Security & Privacy, Computer Graphics (C++)

Certification: CAE-CDE Certificate, NSA/DHS National Center of Academic Excellence in Cyber Defense Education

Honors: CACI C-MAPP Scholar, National Merit Scholar, Dean's List, Eagle Scout, Rotary Youth Leadership Award